

CAT — Communication Assessment Tool

An AI-Assisted Digital Platform for Assessing and Reporting Communication Abilities

SmartAbility 2026 — AI/AR/VR Innovation Challenge | Problem Statement 4 | REC × NIEPMD

Abstract

Speech-language clinicians and special educators routinely assess communication behaviors in individuals with disabilities, but the process today is largely manual: paper checklists, hand-calculated scores, and reports typed from scratch for every case. This is slow, inconsistent across clinicians, and leaves little time for direct intervention. **CAT (Communication Assessment Tool)** is a digital assessment platform that converts this workflow into a structured, guided, and partly automated process, while using generative AI at the exact points where clinicians currently spend the most unproductive time: writing narrative reports and translating clinical findings for families.

Problem

Existing communication assessments rely on static, paper-based checklists that are time-consuming to score, produce reports written in clinical language inaccessible to parents and caregivers, and offer no linked instructional resources to act on the results. This creates a gap between assessment and intervention, and between clinicians and the families they serve.

Proposed Solution

CAT digitizes the entire assessment-to-report pipeline and layers AI on top of it to make the output genuinely useful to both clinicians and caregivers:

- **Digital Present/Absent Checklist** — a structured, editable inventory of communication behaviors administered directly on-screen, replacing the paper form.
- **Automated Scoring Engine** — instantly calculates domain-wise and composite scores as the checklist is completed, eliminating manual tabulation errors.
- **Linked Teaching Videos** — every checklist item is connected to a short instructional video, so a flagged behavior comes with an immediate, actionable teaching resource.
- **Automated Report Generation** — a formatted clinical report is produced the moment scoring is complete.

AI Layer — Where CAT Goes Beyond Digitization

- **AI-Generated Clinical Narrative** — an LLM converts raw domain scores into a coherent, professionally worded clinical summary, saving clinicians the most time-consuming part of reporting.
- **AI-Assisted Scoring from Free-Text Notes** — clinicians can type observations in natural language, and the system suggests Present/Absent scores against the checklist, which the clinician then confirms.
- **Caregiver-Friendly Report Mode** — the same underlying results are re-expressed by the AI in plain, empathetic language for parents, bridging the gap between clinical documentation and family understanding.

Stretch Scope (time permitting)

A session-over-session trend chart to visualize progress, voice-to-text input for faster note-taking during live assessment, and a lightweight browser-based VR module (built with A-Frame) presenting a single 360° scenario with two to three interactive hotspots that feed observations directly into the same scoring engine — demonstrating an immersive, low-friction path toward AR/VR-based assessment without requiring dedicated headset or game-engine development.

Impact

By combining structured digital assessment with targeted generative AI, CAT reduces clinician documentation time, improves consistency of scoring, and — critically — makes results understandable and actionable for the caregivers who support these individuals every day outside the clinic. It is designed to be fully demonstrable within a 10-hour build window while directly addressing the assistive, AI-driven mandate of the SmartAbility Innovation Challenge.

“CAT digitizes the assessment and uses AI to write the clinical narrative, suggest scores from notes, and translate reports for parents versus clinicians.”